

Who's Afraid of Communist AI

Epistemic Transfer and Ideological Convergence in Language Models

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Background: How LLMs Are Aligned

- LLMs are aligned in two stages: SFT, then preference optimization (DPO, RLHF)
 - SFT (supervised fine-tuning) trains on curated instruction-answer pairs
 - Preference optimization further shapes responses from paired comparisons
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- SFT is the most impactful alignment step -- it sets the model's reasoning baseline
 - But we do not understand what SFT transfers beyond its training domain

The Open Question

- Prior work: LLMs carry ideological priors from pretraining (Kronlund 2024, Lee 2026)
 - These studies treat ideology as a static property -- what the model already knows
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- We ask a dynamic question: can targeted SFT shift a model's reasoning?
 - We train on a non-dominant framework: Dialectical Materialism (DM)
 - DM emphasizes structural conditions, systemic contradictions, skepticism of simple causality

DM = Dialectical Materialism (Marxist analytical framework)

Experimental Setup

Training

- Student: Qwen3.5-9B (Instruct variant)
- Teacher: Qwen3.5-27B (generates DM answers only)
- QLoRA with NF4 quantization, rank 32, 7 target modules
- 1,460 DM question-answer pairs, trained 3 epochs
- Single NVIDIA RTX 5090 (32 GB VRAM)

Evaluation

- All benchmarks: bf16 precision, 0-shot greedy decoding
- General capability: MMLU, HumanEval, GPQA Diamond
- Formal causal logic: Corr2Cause (1,162 samples)
- Applied causal reasoning: EconCausal (3,943 samples)

Three Divergent Outcomes

General: Preserved

- MMLU: -0.8pp
(within noise)
- HumanEval: 0.0pp
(identical)
- GPQA: -1.5pp
(within noise)
- No catastrophic forgetting

Formal Causal: +38.3pp

- Corr2Cause: 36.3%
-> 74.6%
- Corrects 520
baseline errors
- Only 75 new errors
introduced
- Net gain: +445
correct answers

Applied Causal: -12.4pp

- Task1 Econ: 60.3%
-> 47.9%
- Task1 Finance:
56.5% -> 43.0%
- Task3: 22.2% ->
11.4%
- All regressions
highly significant

The Hedging Artifact

Dominant Failure Mode: Positive-to-Mixed Hedging

- Finetuned model converts correct positive predictions (+) to ambiguous "mixed"
- Task1 Econ: 96 of 182 regressions are + to mixed (52.7%)
- Task1 Finance: 95 of 174 (54.6%)
- Positive-effect conversion accounts for 77-85% of all Task1 regressions

Why? A Transferred Epistemic Prior

- Hedging is ABSENT from training data -- teacher hedges only 4.0% of the time
- Model internalizes DM principles: "outcomes depend on material conditions"
- It then applies this skepticism indiscriminately -- even where definitive answers exist
- This is not a reasoning error. It is a transferred epistemic prior.

Why Formal Causal Improves While Applied Causal Regresses

Corr2Cause: +38.3pp

Formal structural analysis on causal graphs

- DM training strengthens structural reasoning: trace dependencies, find confounders
- Non-child descendant: 13.0% -> 94.3% (+81.3pp)
- Non-parent ancestor: 14.4% -> 87.2% (+72.8pp)
- Largest gains on the most structurally complex templates

EconCausal: -12.4pp

Applied causal identification from empirical context

- DM skepticism prior overrides empirical directional claims
- Model learned "everything depends on structural conditions"
- Applies this heuristic where the correct answer is a definitive + or -
- Result: systematic under-prediction of positive effects

Causal reasoning has two distinct bottlenecks: formal logic (strengthened by DM) and identification details (weakened by DM). Mirrors findings of Syrgkanis 2026.

Implications for Alignment Research

SFT transfers epistemic priors, not just behavior

- Hedging artifact absent from training data, emerges from internalized framework
- Shift operates at level of reasoning heuristics, not response patterns

Standard alignment may carry unaudited priors

- If DM training produces detectable transfer, RLHF likely does too
- Helpfulness training may transfer deference to authority, controversy avoidance

The Identification-Logic Split as a diagnostic

- Intervention improves formal logic but impairs applied identification?
- Signal of over-generalized structural skepticism

Summary

- Fine-tuned Qwen3.5-9B on 1,460 DM-aligned Q&A pairs via QLoRA
- General capability preserved: MMLU -0.8pp, HumanEval unchanged at 70.7%
- Formal causal reasoning improved dramatically: Corr2Cause +38.3pp
- Applied causal reasoning regressed: EconCausal -12.4pp to -13.5pp
- Regression from single emergent artifact: positive-to-mixed hedging bias
- Hedging absent from training data -- a transferred epistemic prior
- Conclusion: SFT transfers reasoning priors, not just behavioral patterns

Thank You

Questions?

Key Takeaway

SFT transfers epistemic priors, not just behavior.

Alignment training carries hidden reasoning shifts that should be audited.

Ongoing: GRPO with process rewards to break the hedging prior
(dual advantage: outcome correctness + meta-reasoning for commitment)